

STOCK RESEARCH Unified Framework — v2.2 addendum

APPLIED-AND-VERIFIED RELEASE. This document does not replace the source v2.1 PDF (STOCK_RESEARCH_Unified_Framework_v2_1.pdf) — it is a set of findings and binding corrections discovered by actually *implementing* v2.1 as executable code (multibagger/) and running it against six real, cross-sector, hindsight-confirmed multibaggers. Per **M1 Base-Rate Discipline**, every finding below states its denominator; per **M2 Complexity Budget**, each fix is a correction to an existing rule, not a new one, so it does not draw against the complexity budget the way Part 7 did in v2.1.

Read this alongside multibagger/reports/BACKTEST_REPORT.md, which is the evidence base for every finding here.

What v2.2 is, and isn't

v2.1 is a *methodology* — a document a human analyst reads and applies by hand. v2.2's contribution is turning every rule v2.1 states as a formula or numeric threshold into executable code (multibagger/engine.py, gates.py, moat.py, mqf.py, composite.py), covered by 19 unit tests including an exact reproduction of the PDF's own worked example (5x/5yr, 38.0% required return, 68% earnings share — see tests/test_engine.py).

Rules the PDF explicitly leaves to analyst judgement (moat-source ratings, MQF pillar scores, industry-phase classification) are **not** automated — they remain typed inputs a human must supply with Tier 1-3 evidence, exactly as M6 requires. The engine only mechanizes what the PDF itself mechanizes.

Data honesty note (read before trusting any number in the backtest report): live financial-data tooling was unavailable in the build environment — the connected market-data API was gated to symbol search only, and aggregator sites (Screener, Moneycontrol, Tijori, Macrotrends, WSJ) were unreachable. The six backtest dossiers mix a handful of verified, cited anchor facts with Tier-5 (general-knowledge) reconstruction of everything else — explicitly *not* the Tier 1/2 evidence M6 requires for a real gate-clearing verdict. Every finding below that depends on the backtest's *numbers* is flagged as data-uncertain; findings that depend only on the framework's *internal logic* (the gaps in sections 1-3 below) hold regardless of data quality, because they were caught by reading the code's own behaviour against the PDF's text, not by trusting the numbers it output.

1. Gap: G1 has no BFSI/utility carve-out (found applying it to Bajaj Finance)

The rule as written (v2.1 PDF, Part 2): "G1 Leverage — D/E above 2.0x AND interest coverage below 1.5x simultaneously → DISCARD." G2's text explicitly exempts "regulated utilities, BFSI, turnarounds." G1's does not.

What happened when applied literally: Bajaj Finance — one of the best- documented compounders on the NSE, a lend-to-earn NBFC — structurally runs D/E of 4-8x because borrowing at a spread *is* the business model. Applied literally, G1 auto-discards every healthy NBFC before Stage 2 even runs, which is absurd on its face: leverage is the fuel of a lender's ROE, not a distress signal, provided the *spread* and *asset quality* are sound.

Fix (binding, entering at PILOT per M1 — denominator: n=1, this case): gates.py::g1_leverage now exempts is_regulated_utility_or_bfsi names from the D/E test entirely, returning a TEAL informational finding rather than a pass or fail, with an explicit note that a sector-appropriate leverage test (e.g. capital adequacy ratio, gross/net NPA trend, cost-of-funds trend) is needed in its place and is not yet implemented — flagged, not quietly scored as clean.

Also found, same root cause: Section 5e's ROIIC/RR formula defines "invested capital" as net fixed assets + CWIP + net working capital — a manufacturing/services balance-sheet model. For a lender, the capital that compounds is the loan book funded by borrowings; net fixed assets are almost irrelevant. Section 6B's profit-pool/runway gate has the same problem in reverse — there is no product-market TAM for a loan book. composite.py::evaluate now emits an

explicit warning note on any BFSI name that Section 5e's engine-growth figure should not be trusted, and skips the 6B gate's `addressable_profit_pool_cr` requirement (left None) rather than scoring a meaningless number.

Recommended for v2.3, not yet built: a parallel BFSI engine test — incremental spread income ÷ incremental AUM in place of ROIIC, book-value-per-share CAGR in place of RR, credit-cost- adjusted ROE as the floor test in place of the ROIIC-vs-cost-of-capital gate.

2. Gap: Section 5e silently double-counts against Section 6A's own margin-delta term

The rule as written: Section 5e Gate 1 requires " g (ROIIC×RR) ≥ the required net-profit CAGR from the Part 7 N1 return bridge." Separately, Section 6A's operating-leverage-inflection test says: "the first two years of a multibagger are usually margin, not volume... an unevidenced margin delta is set to zero in the bridge and the bridge re-run" — implying the N1 bridge's required-EPS-CAGR figure is meant to be splittable into a revenue-driven term and a margin-delta term, with the latter separately evidenced and gated by 6A.

The gap: as specified, Section 5e's Gate 1 never performs that split — it compares g (which is, by definition, a *volume/reinvestment-driven* growth concept: more capital invested at a given return) against the

full required net-profit CAGR, which legitimately also contains the margin-delta term. That silently demands reinvestment alone deliver 100% of required earnings growth, including the margin contribution 6A already claims to evidence and gate separately. Applied to all six backtest names, this is the single most common blocker (see report) — worth checking whether it is real strictness or a specification gap before trusting the "AVOID" it produces at face value. It is the latter: comparing a pure- volume growth figure against a blended volume+margin target isn't a stricter test, it's an internally inconsistent one.

Fix (PILOT, denominator $n=6$ — this backtest; needs independent confirmation before PROVISIONAL):

`engine.py::compute_return_bridge` now decomposes the bridge: `required_revenue_cagr = (1+required_eps_cagr) / (1+margin_delta_used) - 1`, where `margin_delta_used` is the analyst's assumed `margin_delta_cagr` **only if** `margin_delta_evidenced=True` **and** a `margin_delta_mechanism_note` is actually filled in — otherwise it is zero, per 6A's own rule. Section 5e's Gate 1 (`composite.py::build_gate_table`) now compares g against `required_revenue_cagr`, not the blended figure.

3. Gap: "price vs. the bridge" and "engine strength" were accidentally the same axis

The rule as written (Part 6, the Engine×Price matrix): the matrix is explicitly **two independent axes** — "the primary axis is now the compounding engine (the Part 7 gate); the secondary axis is price versus the return the engine can deliver." The matrix keeps a real cell open for "ENGINE WEAK, price BELOW bridge → value candidate only... never above 3%" — i.e. even a business whose forward engine can't clear an ambitious target can still be a modest position if today's price is undemanding.

The gap (introduced by this implementation, not the PDF — flagged honestly rather than presented as a PDF-derived rule): the first version of `composite.py::price_vs_bridge` reused the *same* engine-growth-vs-N1-target ratio for both axes. That makes them perfectly correlated by construction — a name failing the engine axis by a wide margin will, tautologically, also show "price ABOVE bridge," which deletes the BELOW-bridge value-candidate cell entirely. Running the backtest with this version, every WEAK-engine name landed at IGNORE/AVOID with no differentiation — a red flag that the two axes weren't actually independent.

Fix: price_vs_bridge now scores the price axis against what the business has **already demonstrated** — a Peter Lynch PEG≈1 heuristic (fair P/E ≈ trailing 3-year revenue CAGR in percentage points, bounded to [8, 60]) — rather than against the aspirational N1 target. Re-running the backtest after this fix immediately produced a differentiated result: Bajaj Finance moved from a blanket AVOID to "SPECULATIVE / VALUE PATH ONLY — WATCH," because its entry multiple was undemanding relative to its own historical growth even though its engine didn't clear the aggressive 10x/10y target — exactly the cell the PDF's matrix intends to keep open, and exactly the differentiation a same-axis bug had been silently deleting.

4. Refinement: Section 5e's engine-growth gate graduated (PILOT)

Finding: treating "g < required revenue CAGR" as a single binary hard-fail is inconsistent with how the rest of the suite grades severity — G3 has four states, MQF has four bands, moat trajectory has three. A name missing the requirement by a hair (g at 95% of what's needed) was previously scored identically to one missing it by 70%.

Fix: a near-miss (g at 80-99% of the required revenue-driven CAGR) now soft-fails — it caps the name at Steady Compounder rather than routing to WEAK/IGNORE, matching how an ORANGE moat-trajectory miss is already handled elsewhere in the same table. **Denominator so far: n=0** — none of the six backtest names actually fell in the 80-99% band (they missed by 30-70%), so this refinement is logically motivated by the inconsistency above but has not yet been exercised by a real near-miss case. Flagged PILOT, not claimed as validated.

5. Practical finding: Section 5e's precise inputs are not sourceable from

Tier 3 aggregators, let alone Tier 1/2, through automated tooling

Attempting to source real multi-year CWIP, net-working-capital, and segment-level margin data for six companies via web search and site fetches failed almost completely — every major Indian equity-data aggregator (Screener, Moneycontrol, Tijori, Macrotrends) was unreachable from the build environment, and web search returned only recent-quarter snippets for most names, not historical annual series. This is not a framework defect, but it is a real, discovered barrier to running Section 5e or the N8 funnel screen as a repeatable, automated pipeline: **recommend the framework define an explicit "screening-grade" approximation** — e.g. invested capital ≈ capital employed (net worth + total debt – cash), the same denominator already used for ROCE — for use only in the N8 funnel screen and initial triage, with a hard rule (already implied by M6, now made explicit) that no BUY/ACCUMULATE verdict may cite this approximation as its Section 5e evidence; a real verdict re-derives ROIIC from the precise definition using Tier 1/2 statements.

6. Headline empirical finding from the backtest (read with the data caveat above)

All six real, hindsight-confirmed multibaggers — Astral (Industrials), Titan (Consumer), Divi's Labs (Pharma), Persistent Systems (IT), Bajaj Finance (Financials), Dixon Technologies (Consumer Electronics) — evaluated as of a snapshot date *before* their known multi-year re-rating, land at Engine verdict WEAK and fail to reach Multibagger Candidate tier under this engine, even after the three fixes above. In five of six cases the sole or dominant blocker is the Section 5e engine-growth gate, missing the required revenue-driven CAGR by 30-70%, not by a hair.

Two readings are both plausible and neither should be dismissed:

- **(a) Data reading:** the Tier-5 reconstructions likely understate true

historical ROIC/reinvestment for all six names, since real balance-sheet detail wasn't obtainable. This is the more likely primary explanation and is exactly why M6 exists — do not treat any single number here as a verdict on these six real companies.

- **(b) Framework reading:** even granting generous, real, hindsight-

informed assumptions (target multiples and horizons were set to roughly match what each stock *actually* delivered, not chosen aggressively), a meaningful share of these six real returns came from multiple re-rating beyond what a conservative, percentile-capped N1-2 exit-multiple ceiling would sanction in advance. That is very likely a real, not an artifactual, property of the framework — and it is not obviously a flaw: v2.1's own "failure mode" section already names this exact tension in the abstract ("a framework this conservative on governance will systematically exclude some genuine multibaggers... that exclusion is deliberate and correct"). This backtest gives that abstract warning a concrete, quantified illustration across six sectors rather than a single anecdote.

Recommendation: track this at the next M1 review the same way S1/S12 tracks its own false-negative risk — log, for every name that fails only the Section 5e/N1 gates while clearing everything else (moat, industry phase, governance, variant perception), what actually happened to it over the following 3-5 years. If a pattern of missed compounders concentrated on this specific gate emerges with a real denominator, that is the evidence needed to revisit the gate's strictness under M1 — not a hunch, a Tier-5 backtest, or this document alone.

M2 accounting for v2.2

Every change above is a **correction to an existing v2.1 rule's implementation**, not a new rule: G1's carve-out extends G2's existing carve-out logic to a rule that should have had it from the start; the 5e/6A split makes two existing sections consistent with each other rather than adding a new gate; the price-vs-bridge fix repairs an implementation bug introduced by this codebase, not a PDF rule; the graduated engine-growth band reuses the multi-state pattern G3 and MQF already established. No new rule is added, so v2.2 draws nothing against the M2 complexity budget.

Denominator ledger (M1)

Finding	Status	Denominator
G1 BFSI carve-out	PILOT	n=1 (Bajaj Finance)
5e/6A margin-delta split	PILOT	n=6 (this backtest)
Price-vs-bridge axis decoupling	PILOT	n=6 (this backtest)
Engine-growth graduated band	PILOT	n=0 (not yet exercised by a real near-miss)
Screening-grade 5e approximation	Proposed, not implemented	n=0
BFSI-variant engine test	Proposed, not implemented	n=0

None of the above may be described as VALIDATED or PROVISIONAL in any report generated by this engine until a real, out-of-sample denominator exists — consistent with M1's own text, which already applies this standard to every Part 7 rule in v2.1.